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SDGsR: An R package for interacting with the UN Sustainable Development Goals

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Software

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Summary

The UN World Data Forum highlighted that the power of data can only be harnessed, and its true value realized, by bringing it together, through statistical analysis, and by connecting it to other data sets to inform solutions for challenges facing humanity (Sustainable Development, 2023) .The R programming language is the most utilized programming language for statistical analysis (Computer Science, 2023). Because of R's open-source nature, its robust community of users, and its support for reproducibility, it has quickly become a key tool in the Open Science movement. R facilitates researchers' and data scientists' abilities to disseminate state-of-the-art statistical methods, share their own analysis work, and reproduce the work of others (Helwig, 2024). These characteristics of R make it an optimal tool for statistical analysis of the SDG's, however, no current tool exists to leverage the United Nations Statistics Division SDG API and seamlessly import SDG data into R. The SDGsR package addresses this gap, providing a valuable resource for integrating SDG data with R for more comprehensive analysis and insights.

Statement of need

The Sustainable Development Goals (SDGs) are a set of measurable, specific, and time-sensitive goals built with a mission for national development. Available, accurate, and quality data and statistics are crucial in making significant progress toward achieving these goals (Nilashi et al., 2023). Furthermore, analyzing SDG data is necessary for assessing progress, informing policy-making, allocating resources, and identifying existing gaps (RELX, n.d.).

R is an open-source software and a powerful programming language commonly used for data analysis, modeling, and visualization. It allows its users to utilize data to perform analysis, produce visualizations, and promote data reproducibility and transparency in their work (Lai et al., 2023). R packages are fundamental units of reproducible R code that often include R functions, documentation, and sample data. Organizing code in a package not only assists in running the user's code more efficiently, but also provides a space to share their code with others where they can easily download and run code (Wickham & Bryan, 2023). Utilizing R and R packages to analyze SDG data is becoming increasingly more common in the data analysis environment. Other existing R packages include SDGdetector (Li et al., 2023) and text2sdg (Meier et al., 2023).

Application programming interfaces (APIs) are useful for data integration, enrichment, access, and sharing. APIs functionality for exchanging information have a significant impact on growth, efficiency, and innovation in a variety of fields such as healthcare, automotive, and banking. The potential for APIs has been recognized through the 21st Century Cures Act where it is required that health information technology developers make standards-based APIs available (Gordon & Rudin, 2022).

The United Nations Statistics Division has an API that allows users to explore official SDG data reported by custodian agencies (https://unstats.un.org/sdgs/UNSDGAPIV5/swagger/). However, manually downloading the most updated data from the SDG API comes with the additional steps of changing the data format, reading the file into R, etc. SDGsR addresses this challenge by leveraging the SDG API in an R package with functions to access updated SDG data efficiently in the R environment. Such data and information are pulled directly from SDG API v5.

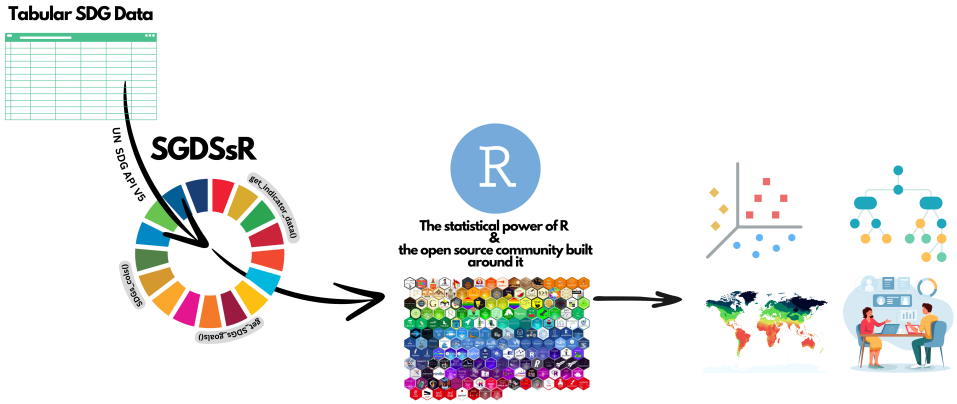

Figure 1: SDG Goals Table

Figure 1. Flowchart demonstrating data flow from the UN Sustainable Development Goals API through SDGsR to leverage the statistical power and open source community of R.

Functionality

SDGsR is an R package developed to interact with and leverage the UN Sustainable Development Goals API (UNSDGAPIV5) by providing the ability to: - extract textual information on each SDG, including targets and indicators - identify country codes - utilize the SDG color palette with ggplot - pull most recent/historical indicator data for specific countries

The package is inspired by an article discussing extracting API data in R (Kim, 2019). SDGsR is available on GitHub at https://github.com/DrMattG/SDGsR.

Usage

To begin using this package, a user can use the following installation code:

```
devtools::install_github("DrMattG/SDGsR", dependencies = TRUE)
```

```
# load package
library(SDGsR)
```

To get the text of the Goals, use the SDGsR::get_SDGs_goals() function. This gathers together all the Goals in to a single dataframe.

```
Goals <- get_SDGs_goals() # get goals, targets, and indicator descriptions as a datafram
```

```
Goals %>%
  select(goal, code, description) %>%
  filter(goal == 15) %>%
  kableExtra::kable() %>%
  kableExtra::kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

goal	code	description
15	15.1	By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements
15	15.2	By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally
15	15.3	By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world
15	15.4	By 2030, ensure the conservation of mountain ecosystems, including their biodiversity, in order to enhance their capacity to provide benefits that are essential for sustainable development
15	15.5	Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species
15	15.6	Promote fair and equitable sharing of the benefits arising from the utilization of genetic resources and promote appropriate access to such resources, as internationally agreed
15	15.7	Take urgent action to end poaching and trafficking of protected species of flora and fauna and address both demand and supply of illegal wildlife products
15	15.8	By 2020, introduce measures to prevent the introduction and significantly reduce the impact of invasive alien species on land and water ecosystems and control or eradicate the priority species
15	15.9	By 2020, integrate ecosystem and biodiversity values into national and local planning, development processes, poverty reduction strategies and accounts
15	15.a	Mobilize and significantly increase financial resources from all sources to conserve and sustainably use biodiversity and ecosystems
15	15.b	Mobilize significant resources from all sources and at all levels to finance sustainable forest management and provide adequate incentives to developing countries to advance such management, including for conservation and reforestation
15	15.c	Enhance global support for efforts to combat poaching and trafficking of protected species, including by increasing the capacity of local communities to pursue sustainable livelihood opportunities

Figure 2: SDG Goals Table

The most useful function is the `SDGsR::get_indicator()` function that allows the user to query the data for a specific indicator in a specific country. First, it is necessary to know the correct code for the country and for the indicator as the API uses these to distinguish which data to export. There is an country list in the package that can be utilized to find country names.

```
head(get_country_list())
#> [1] "Afghanistan"    "Åland Islands"  "Albania"        "Algeria"
#> [5] "American Samoa" "Andorra"
```

To find a specific indicator, a user must know the right code to use. Users can retrieve a list of all the indicators by using `SDGsR::get_indicator_list()`. This example demonstrated indicators for Goal 15 Life on Land.

```
ind_lst <- get_indicator_list()

ind_lst %>%
  filter(goal=="15") %>%
```

```
  select(target, code, description) %>%
  kableExtra::kable()
```

target	code	description
15.1	15.1.1	Forest area as a proportion of total land area
15.1	15.1.2	Proportion of important sites for terrestrial and freshwater biodiversity that are covered by protected areas, by ecosystem type
15.2	15.2.1	Progress towards sustainable forest management
15.3	15.3.1	Proportion of land that is degraded over total land area
15.4	15.4.1	Coverage by protected areas of important sites for mountain biodiversity
15.4	15.4.2	(a) Mountain Green Cover Index and (b) proportion of degraded mountain land
15.5	15.5.1	Red List Index
15.6	15.6.1	Number of countries that have adopted legislative, administrative and policy frameworks to ensure fair and equitable sharing of benefits
15.7	15.7.1	Proportion of traded wildlife that was poached or illicitly trafficked
15.8	15.8.1	Proportion of countries adopting relevant national legislation and adequately resourcing the prevention or control of invasive alien species
15.9	15.9.1	(a) Number of countries that have established national targets in accordance with or similar to Aichi Biodiversity Target 2 of the Strategic Plan for Biodiversity 2011–2020 in their national biodiversity strategy and action plans and the progress reported towards these targets; and (b) integration of biodiversity into national accounting and reporting systems, defined as implementation of the System of Environmental-Economic Accounting
15.a	15.a.1	(a) Official development assistance on conservation and sustainable use of biodiversity; and (b) revenue generated and finance mobilized from biodiversity-relevant economic instruments
15.b	15.b.1	(a) Official development assistance on conservation and sustainable use of biodiversity; and (b) revenue generated and finance mobilized from biodiversity-relevant economic instruments
15.c	15.c.1	Proportion of traded wildlife that was poached or illicitly trafficked

Figure 3: SDG Goals Table

This example looks at the indicator for Target 15.4, "15.4.1: Coverage by protected areas of important sites for mountain biodiversity" and specifies Norway as our country of interest. A user can confirm that Norway is named Norway on our country list, however, there are some countries named in different ways than what might be commonly expected, e.g. Vietnam is specified as "Viet Nam", Venezuela is specified as "Venezuela (Bolivarian Republic of)").

```
Norway_code <- lookup_country(code="M49", country = "Norway")
Norway_code
#> [1] 578
```

```
Norway <-get_indicator(Country = Norway_code, indicator = "15.4.1")
```

A user can utilize the SDGsR::SDGs_colours() function in conjunction with ggplot to match plot colors with the corresponding SDG color.

```
Norway %>%
  select(timePeriodStart, value, seriesDescription) %>%
  ggplot(aes(timePeriodStart, as.numeric(value))) +
  geom_area(fill = SDGs_cols("Goal15"), color = "black") +
  labs(x = "Year", y = "% Mountain Area Protected") +
  ggtitle(label = paste0(Norway$seriesDescription[1])) +
  ggpubr::theme_pubclean() +
  theme(plot.title = element_text(size = 12))
```

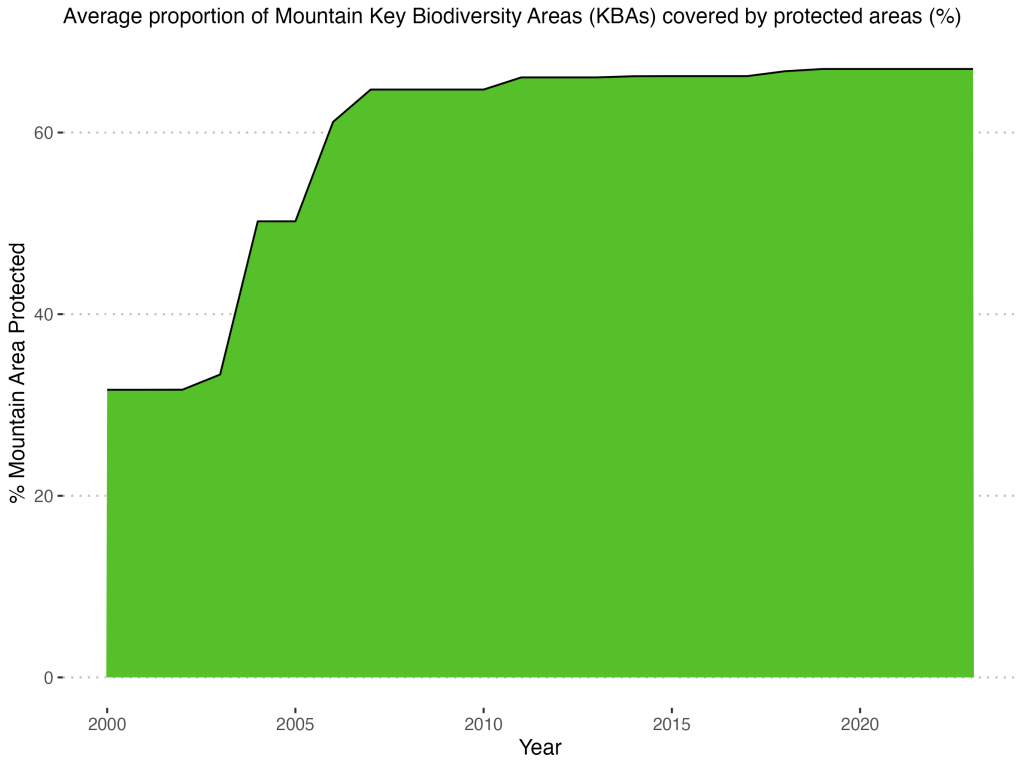

Figure 4: SDG Goals Table

Acknowledgements

We acknowledge contributions from….

References

Computer Science, R. U. D. of. (2023). *12 best programming languages for data science and analytics*. https://cs.rice.edu.

Gordon, W. J., & Rudin, R. S. (2022). Why APIs? Anticipated value, barriers, and opportunities for standards-based application programming interfaces in healthcare: Perspectives of US thought leaders. *JAMIA Open*, *5*(2). https://doi.org/https://doi.org/10.1093/jamiaopen/ooac023

Helwig, N. (2024). *Investigating the r in (r)evolution of open science*. Department of Psychology and School of Statistics, University of Minnesota, https://cla.umn.edu/psychology/news-events/story/investigating-r-revolution-open-science.

Kim, A. M. H. (2019). *Getting SDG data using APIs*. https://rpubs.com/angelamhkim/sdgapi.

Lai, J., Cui, D., Zhu, W., & Mao, L. (2023). The use of r and r packages in biodiversity conservation research. *Diversity*, *15*(12). https://doi.org/https://doi.org/10.3390/d15121202

Li, Y., Frans, V. F., Song, Y., Cai, M., Zhang, Y., & Liu, J. (2023). *SDGdetector: Detecting sustainable development goals and targets in text* (Version 2.7.3). https://doi.org/10.32614/CRAN.package.SDGdetector

Meier, D. S., Mata, R., & Wulff, D. U. (2023). *text2sdg: Detecting UN sustainable development goals in text* (Version 1.1.1). https://doi.org/10.32614/CRAN.package.text2sdg

Nilashi, M., Boon, O. K., Tan, G., Lin, B., & Abumalloh, R. (2023). Critical data challenges in measuring the performance of sustainable development goals: Solutions and the role of big-data analytics. *Harvard Data Science Review*, *5*(3). https://doi.org/https://doi.org/10.1162/99608f92.545db2cf

RELX. (n.d.). *Data & analytics*. RELX SDG Resource Centre, https://sdgresources.relx.com/data-analytics.

Sustainable Development, I. I. for. (2023). *UN world data forum bulletin: Summary of the UN fourth world data forum, 24-27 april 2023*. https://enb.iisd.org/un-world-data-forum-2023.

Wickham, H., & Bryan, J. (2023). *R packages: Organize, test, document, and share your code* (eBook). O'Reilly Media. https://r-pkgs.org/

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